

Solution Requirements

Date	04 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely access the application using login/session management.</i>	Medium
2	Authorization levels	<i>Different access levels can be provided for students and administrators.</i>	Low
3	External interfaces	<i>System integrates AI models like Whisper for speech-to-text and NLP models for concept analysis.</i>	High

4	Transactions processing	<i>Handles audio upload, speech conversion, analysis processing, and result generation.</i>	High
5	Reporting	<i>Generates performance scores, feedback reports, charts, and downloadable PDF reports.</i>	High
6	Business rules, etc.	<i>Evaluates user explanation based on semantic similarity, speech quality, and concept accuracy.</i>	High
7	Compliance to laws or regulations	<i>Maintains user data privacy and secure handling of uploaded audio files.</i>	Medium
8	<i>Other</i>	<i>Provides a user-friendly Streamlit interface for voice-based learning evaluation.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>System should process uploaded audio and generate results quickly.</i>	<i>Analysis completed within few seconds</i>
2	Scalability	<i>Application should support multiple users and future feature expansion.</i>	<i>Supports increasing users and data</i>
3	Security & Data Privacy	<i>Protects uploaded audio files and generated reports.</i>	<i>Secure storage and data handling</i>
4	Reliability & Availability	<i>Application should be accessible without frequent failures.</i>	<i>Stable deployment on Streamlit Cloud</i>
5	Usability & Accessibility	<i>Interface should be simple and easy for students to use.</i>	<i>Clean UI with easy navigation</i>
6	Other	<i>Application should be maintainable and easy to upgrade.</i>	Modular code structure